preventing or at least minimising wasteful use of precious resources.

3. India, situated in the monsoon belt of south-east Asia, is even now dependent to a large extent, on rains for its agriculture. Rainfall is unevenly distributed, and in some years the monsoon fails, resulting in drought. It is a measure of the resilience injected into our agriculture by S&T inputs that we are not so severely affected by droughts now as in the past.

   The geographic situation of the country forces us to learn to cope with inconsistent monsoons. The average annual rainfall we receive is about 370 million hectare-metres of which about 80 million hectare- metres of water seeps into the soil.

4. There was a time when the multipurpose river valley projects were hailed as the harbingers of prosperity. Not anymore. These “temples of Modern India” as Jawaharlal Nehru termed the dams, have no doubt proved an important factor in achieving a successful breakthrough in agriculture and a measure of selfsufficiency in food production.

   However, their success has been overrated. Serious problems have arisen. Heavy siltation of the reservoirs of major dams reduces storage capacity, and they become incapable of absorbing heavy floods. A disastrous fallout is the degradation of the soil in the command areas of irrigation projects due to waterlogging and soil salinity.

5. Sprinkler irrigation is water-efficient, and has been introduced in canal irrigated areas of Haryana, Rajasthan and Madhya Pradesh. Drip irrigation is being encouraged in Maharashtra, Kamataka, Andhra Pradesh, Orissa, and Tamil Nadu for fruit crops. ‘Fertigation’ is an innovation: the application of fertiliser through drip irrigation to maximise farm productivity with available water.

6. Agriculture involves deliberate effort to modify the earth’s surface through cultivation of crops and rearing of livestock for sustenance or economic gain. Modern agriculture has resulted in over-cropping (causing soil deterioration), overgrazing (destroying grass cover), and deforestation (loss of green cover)—all affecting the environmental pattern.